BAAVANSH REDDY GUNDLAPALLI

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SUMMARY

Computer Science graduate from Rutgers University (Magna Cum Laude, 3.76 GPA) targeting backend and AI/ML engineering roles. Builds REST APIs, authentication systems, and data pipelines, with applied AI work in retrieval and embeddings — including AgentMem, an open-source memory library for AI agents. Backed by a strong systems foundation across CPU simulators, compilers, and operating systems, built in C and Python. Personal portfolio includes an AI assistant recruiters can chat with to ask about this work.

EDUCATION

Rutgers University — New Brunswick, NJ May 2026

B.S. in Computer Science, Magna Cum Laude | GPA: 3.76 / 4.0 | Dean's List (4 semesters)

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems, Compilers, Software Engineering, Databases, Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL

Frameworks & Web: Node.js, Express, FastAPI, React, Next.js, REST API design

Databases & Storage: PostgreSQL, MySQL, SQLite, MinIO (S3-compatible object storage)

AI & ML: RAG pipelines, vector search, embeddings, semantic search, PyTorch, NumPy, prompt engineering, LLM evaluation

Tools & Concepts: Git, Linux, unit testing (pytest, JUnit), object-oriented design, system design, Agile collaboration

EXPERIENCE

Backend Developer Intern · Velarro AI Research Center (Metasys Global) Jan 2026 – Feb 2026

- Built the document-ingestion layer for an AI semantic-retrieval system, integrating MinIO (S3-compatible) object storage to programmatically extract heterogeneous, mixed-format files.
- Engineered the file-processing workflow that staged and prepared extracted documents for downstream chunking, embedding, and vector search.

Full-Stack Engineering Extern · Fidelis — LIVEY Event Platform Jan 2026 – May 2026

- Co-built LIVEY, a full-stack event-discovery platform, by developing React search and filtering components and engineering REST API routes across the front and back end.
- Integrated Supabase (PostgreSQL) for real-time data storage, user authentication, and event management across the platform.

Software Engineering Extern · Little Red Riding Hood Inc. Sep 2025 – Dec 2025

- Engineered secure authentication for a safety-focused rideshare platform, implementing role-based login and credential management for separate rider and driver accounts.
- Built driver session tracking and real-time geolocation handling that supported dispatch coordination and passenger-safety workflows.

Python Programming Intern · InternPe Sep 2024 – Oct 2024

- Developed Python automation scripts for data processing and file handling, applying object-oriented design principles across structured engineering assignments.

PROJECTS

AgentMem — Open-Source Memory Library for AI Agents | *Python, SQLite, sentence-transformers, pytest* github.com/BaavanshReddy/agentmem

- Built and open-sourced AgentMem, a ~500-line Python library that gives AI agents persistent memory using local SQLite storage and on-device embeddings, with no cloud database or API keys required.
- Implemented semantic retrieval with cosine similarity and a Maximal Marginal Relevance (MMR) diversity mode, plus per-agent namespacing, and verified the public API with 15 passing unit tests.

Chat-with-Baavansh — AI Portfolio with Embedded RAG Assistant | *Next.js, TypeScript, Claude API* baavansh-portfolio.vercel.app

- Built a personal portfolio site with an embedded RAG assistant that answers recruiter questions from a structured knowledge base, pairing a Claude-powered engine with an automatic in-browser fallback.

RISC-V CPU Simulator | *C, Computer Architecture* github.com/BaavanshReddy/risc-v-simulator

- Implemented a single-cycle RISC-V CPU simulator with full instruction decode and datapath execution, backed by a 1KiB direct-mapped cache with hit/miss tracking to model a realistic memory hierarchy.

TinyL Compiler | *Python, Compilers* github.com/BaavanshReddy/tinyl-compiler

- Built a complete compiler — lexer, recursive-descent parser, AST, intermediate representation, and code generation — translating the TinyL language from a formal grammar to stack-based bytecode.

RuPizza — Object-Oriented Ordering Application | *Java, JavaFX, JUnit* github.com/BaavanshReddy/RUPizza

- Developed a JavaFX desktop ordering application designed with object-oriented principles — inheritance, polymorphism, and the Factory pattern — featuring an order builder, store-order management, and JUnit test coverage.

Neural Network from Scratch | *Python, NumPy, PyTorch* github.com/BaavanshReddy/llm-digit-face-training

- Implemented a perceptron and a 3-layer neural network with hand-coded forward propagation and backpropagation, reaching 89.1% face-recognition and 86.8% digit-classification accuracy validated against a PyTorch baseline.

RESEARCH

LLM FactCheck — Factual QA Evaluation Framework | *Python, RAG, BM25, TriviaQA*

- Co-authored a research study benchmarking LLM factual question-answering across direct prompting, BM25 retrieval, and RAG on a controlled 100-question TriviaQA evaluation set.
- Found that BM25 surfaced supporting evidence in 90% of cases while RAG improved exact-match accuracy by only 3 points (0.66 to 0.69), isolating generation quality — not retrieval — as the bottleneck.

LEADERSHIP & ACTIVITIES

- **Community Service Officer, Rutgers University Police Department** (Jan 2025 – May 2026) — Supported incident response and crowd management for large-scale university events.
- **Leadership Roles, Alpha Phi Delta Fraternity** (May 2024 – May 2026) — Held three elected positions (Risk Management Chair, Head of Judiciary Board, Co-Brotherhood Chair); led safety compliance, conduct governance, and member engagement.
- **Administration Secretary, ENIGMA Technical Society** (Aug 2023 – May 2024) — Organized technical workshops and streamlined communication across society events.

ADDITIONAL

Languages: English, Hindi, Telugu **Activities:** Former NCAA Division I Crew Athlete, Rutgers University